

Matt Love

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SUMMARY

Senior Software Engineer with 12 years of experience building distributed systems across low-level systems programming and modern backend SaaS engineering. Background spans enterprise storage infrastructure in C, high-throughput event pipelines, latency-critical delivery services, platform engineering, and AI-integrated backend systems in Go on GCP. Consistent track record of end-to-end ownership, technical leadership, and delivering systems that other teams depend on.

SKILLS

Languages	Go (5 yrs), C/C89 (7 yrs), Python, Bash, JavaScript
Infrastructure	GCP (Pub/Sub, Datastore/Firestore, GCS, Memorystore, Vertex AI, GKE, Cloud Composer), Kubernetes, Redis
Data & Pipelines	High-throughput event ingestion, distributed async processing, batch and streaming architectures, multi-tier caching, workflow orchestration (Airflow)
AI Systems	LLM API integration, RAG system design, vector embeddings and ANN search (Vertex AI), workflow state management
Practices	Platform engineering, pipeline engineering, SLI/SLO design, incident command, performance profiling (pprof), distributed systems design, Agile/Scrum

EXPERIENCE

Senior Software Engineer · [Pendo](#) 2021 – 2026 | Raleigh, NC (Hybrid)

Platform Engineering | 2025 – 2026

- Owned Pendo's core event ingestion pipeline and jobs framework, the async processing backbone used by every team across the platform. Built on Google Pub/Sub, Datastore, and GCS with a pull-based jobs server architecture supporting parent/child task hierarchies and retry/ack semantics.
- Designed and implemented an automated data migration system using Google Cloud Composer (managed Airflow), reducing a 5-6 hour human-guided process requiring manual API calls, cloud commands, and storage operations to a fully automated, configuration-driven DAG pipeline. Used Claude Code throughout the full development lifecycle from generating a technical spike and breaking work into stories, to implementing the DAGs from documentation, focusing my own time on architecture, validation, and testing.
- Automated a distributed data correction workflow by building a durable FSM on top of a periodic job driver and a Redis-backed state entity. Replaced 5 manual endpoint calls and opaque log monitoring with a self-documenting workflow and real-time UI visibility.
- Served as Tier 1 on-call engineer responding to error, latency, and cost budget alerts, performing Kubernetes remediation, and using BigQuery log analysis for rapid incident triage across distributed services.
- Profiled and optimized a Go event categorization algorithm using pprof against live production data. Found excessive struct copies in deeply nested map iteration, eliminated the redundant iteration, and resolved a chronic SLA breach for a high-volume customer while improving throughput platform-wide.
- Served as incident commander for a company-wide SEV0 that wiped guide content from roughly 100% of public guides across all customer subscriptions. Diagnosed the root cause, led multi-day recovery across three engineering teams, and developed a dual-sort-order version history algorithm to handle edge-case guides. Built post-mortem tooling enabling precise per-customer state reconstruction.
- Built Postboy, a programmatic REST automation platform adopted across the engineering org. Extended GoLand's HTTP client with a Go-based scripting engine, a Chrome extension that automatically synced JWT session tokens and page-context variables to local env files, and a composable IO/tasks library enabling parallelized pipeline workflows. Won the CDO Innovation Award for a companion AST-based tool that auto-generated request files for every backend route by statically analyzing the monorepo.

Guides Backend | 2023 – 2025

- Became the sole backend engineer across two full frontend teams within one year and served as the de facto subject matter expert for all guides backend systems. The most critical was the guide delivery service, the platform's most trafficked endpoint, which required near-zero response times, an embedded real-time eligibility engine that evaluated each visitor's behavioral data against guide targeting rules, goroutine-parallelized evaluation across all candidate guides, and a three-tier cache spanning local memory, Memorystore, and Datastore.
- Resolved a multi-year chronic bug causing translated guide content to return 403 errors across customer apps. Systematically eliminated four distinct failure modes including timeout-induced partial writes, unnecessary retranslation, incorrect write ordering, and a race condition creating mismatched storage object references. Proactively

audited the entire platform, found roughly 80,000 missing files across 10,000 guides, and repaired all of them with zero reprocessing.

- Designed and shipped A/B guide experiments with controlled rollout using hash-based deterministic visitor assignment. Used atomic transactions to keep shared experiment state synchronized between variants.
- Built Guide AI Localization as a hackathon project that became a near-universally adopted product feature. Automated the translation workflow by integrating the Google Cloud Translation API into the existing import and export pipeline and adding an in-app editor so customers never had to manually edit XML files.
- Built a production RAG system integrating customer knowledge bases with guide content generation. Implemented FSM-driven async index lifecycle management and ANN-based semantic retrieval to supplement LLM prompts with relevant context.

Analytics Backend | 2022 – 2023

- Implemented platform-wide referential integrity across a distributed system, preventing deletion of shared resources such as user segments and metadata fields that were referenced across multiple product surfaces including analytics reports, messaging rules, and embedded definitions.
- Led backend development of a comprehensive analytics reporting tool, owning report definition persistence, saved segment management, and integration with the aggregations platform.

Enterprise & Product-Led Growth | 2021 – 2022

- Primary backend engineer for a full SCIM implementation integrated with Okta, enabling enterprise customers to automate user provisioning. Built the external Groups and Users APIs and the internal layer mapping identity provider groups to the platform's hierarchical access control system.
- Delivered a free trial product end-to-end including route guard tier enforcement, an async downgrade job, and graceful artifact deactivation on subscription expiry.

Senior Software Engineer · EMC / Dell EMC (Data Domain) 2014 – 2021 | Research Triangle Park, NC

Seven years building enterprise storage OS software in C89 on a mature, high-reliability codebase with tens of millions of lines of code.

- Delivered NFSv4 from greenfield to production as a core team member. The project included a 4x concurrent client scaling effort (2K to 8K connections) achieved by replacing a global-mutex linked list with a per-entry-mutex hash table in the export authentication hot path. Also served as branch keeper for the multi-year feature branch, maintaining merge hygiene across the entire codebase.
- Developed a cross-platform SDK shipping across Linux, AIX, HP-UX, Solaris, and Windows. Gained deep expertise managing platform-dependent code paths, per-OS build systems, and preprocessor macros across five operating systems.
- Implemented a hot-swappable binary update system enabling seamless patches to tens of thousands of backup clients without reinstallation. Used a wrapper and implementation two-binary architecture with server-side version checking and dynamic linking. The architecture was filed for patent by the principal architect.
- Built developer productivity tools that spread organically across the engineering team at both companies. At Dell EMC, built a multiplexed SSH session manager with a persistent daemon and broadcast command execution across multiple machines. At Pendo, built the REST automation platform described above.

EDUCATION

B.S. Computer Science · North Carolina State University 2014 | Raleigh, NC